

DEVELOPMENTAL DISTURBANCES OF SHAPE AND SIZE OF TEETH

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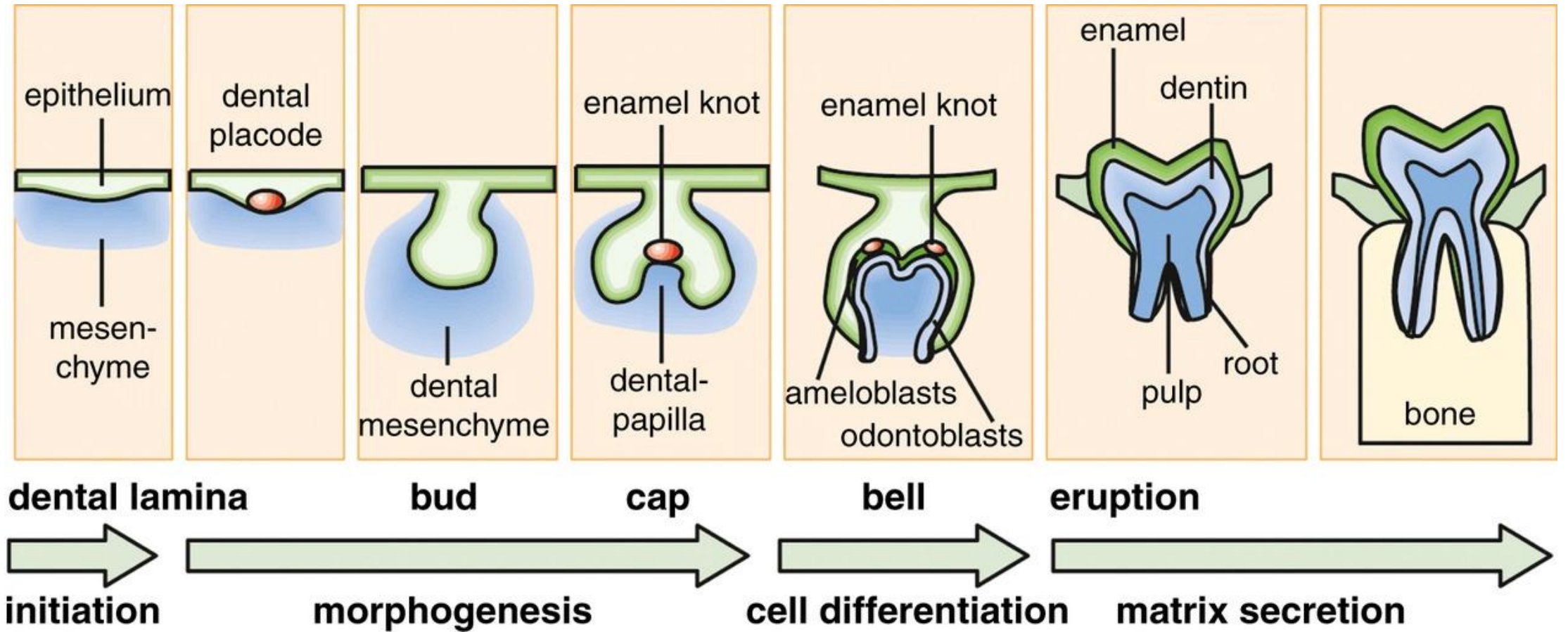
ORAL AND MAXILLOFACIAL PATHOLOGY

LLO: Developmental Diseases affecting teeth

CLO	LLO	By the end of this lecture, students should be able to:	
1.1	L1	Discuss the classifications of the developmental diseases affecting Teeth.	
	L2	Describe-clinical and radiographic features of developmental diseases affecting number, structure, shape and size of teeth	

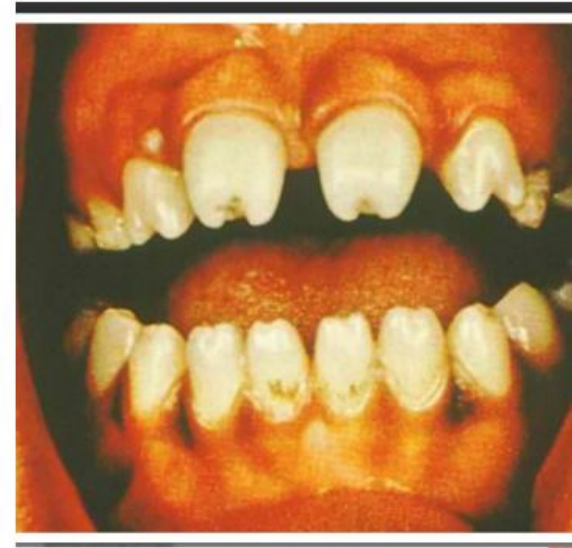
Essential reading:

- Regezi & Sciubba: Oral Pathology Clinicopathologic Correlation; 2017 Elsevier (7th edn),
Chapter 16 – Abnormalities of teeth



Developmental disturbances:

- (1) Size
- (2) Number and Eruption
- (3) Shape/Form
- (4) Defects of Enamel and Dentin



D/Disturbances affecting size of teeth teeth



MACRODONTIA



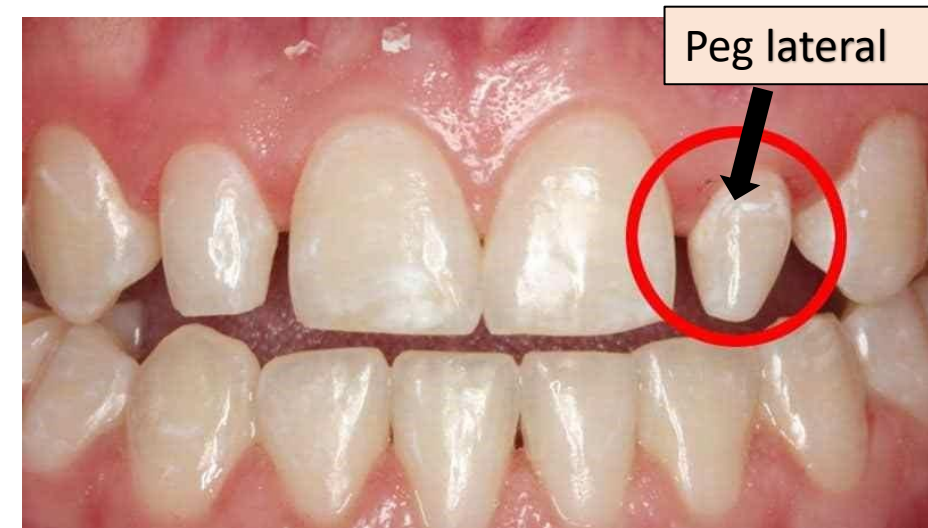
MICRODONTIA

MICRODONTIA (MICRO-SMALL, DONTIA-TEETH)

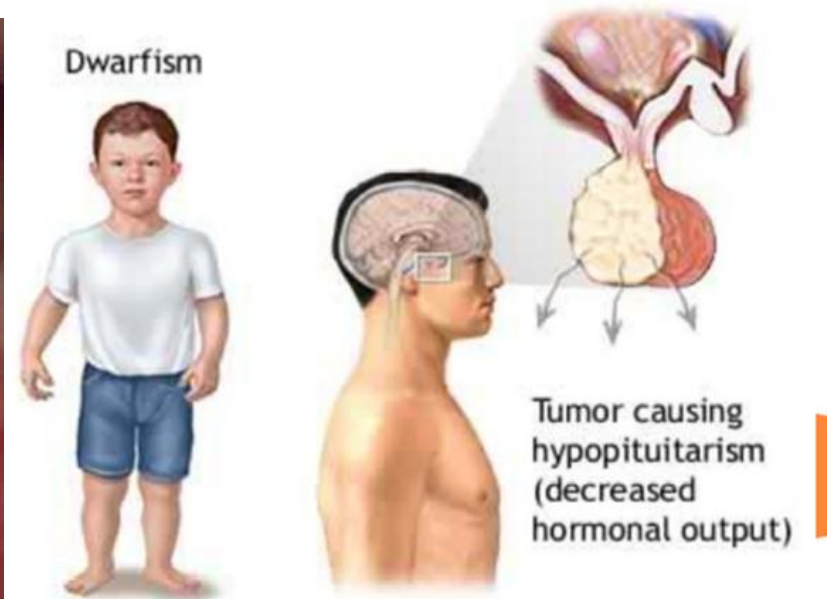
It means teeth that are **smaller** than normal.

TYPES:

1. **True Generalized microdontia:** all teeth affected. As in Pituitary dwarfism
2. **Relative generalized (pseudo) microdontia**
3. **Localized Microdontia:** limited to just one or two teeth. Most common with maxillary lateral incisors (Peg Lateral). The second most commonly seen microdont is the maxillary third molar, followed by supernumerary teeth



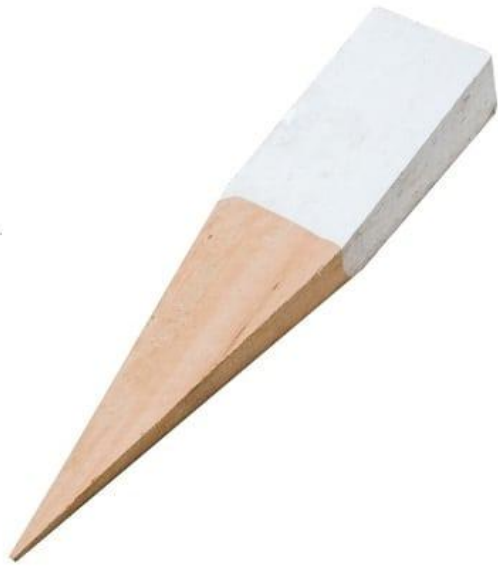
True Generalized microdontia:



Relative generalized (pseudo) microdontia



Localized Microdontia:



Peg lateral incisor: sides converge towards incisally and form cone shaped incisor

MACRODONTIA

- It refers to teeth that are **larger** than normal.
1. True Generalized macrodontia (pituitary gigantism)
 2. Relative Generalized macrodontia (pseudo macrodontia)
 3. Localized macrodontia
- The term '**megadontia**' is also used for this condition.



True Generalized macrodontia (pituitary gigantism)



Relative Generalized macrodontia (pseudo macrodontia)



Localized macrodontia



DEVELOPMENTAL DISEASES AFFECTING SHAPE OF TEETH

GEMINATION

TWINNING

FUSION

CONCRESCENCE

One Tooth



One Bud

Two Teeth



One Bud

Dentin Union
(two teeth)



Two Buds

Cementum Union
(two teeth)



Two Buds

"FUSION"

- "Fusion" is the union of **two tooth germs** during their **development** Resulting in a single large tooth structure.
- Union before calcification: complete fusion
- Cause unknown (trauma)
- Clinical problem: spacing, and periodontal conditions
- Fusion presents clinically as a single enlarged tooth with one tooth missing on counting.



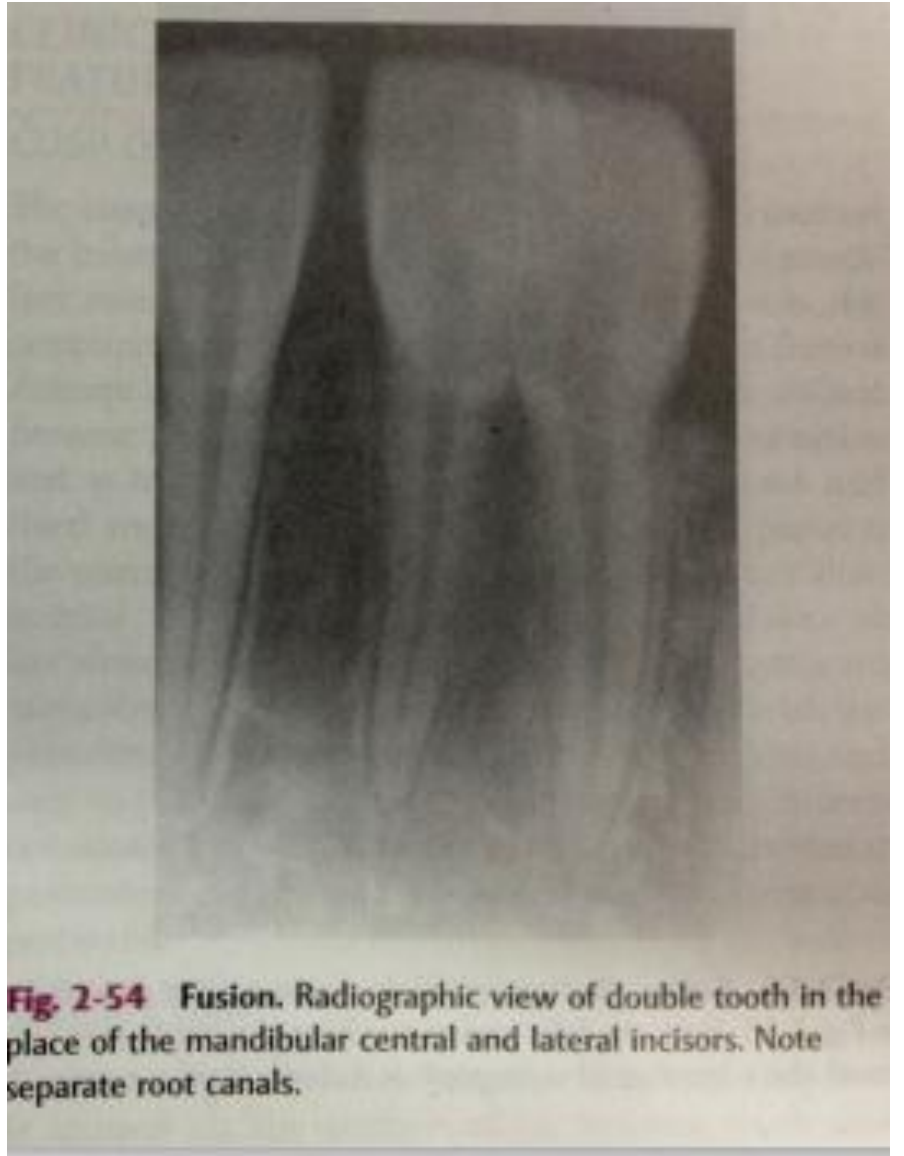


Fig. 2-54 Fusion. Radiographic view of double tooth in the place of the mandibular central and lateral incisors. Note separate root canals.

GEMINATION

- Geminated teeth are anomalies which arise from an attempt at division of a **single tooth germ** by an invagination, with resultant incomplete formation of two teeth.
- The structure is usually one with two completely or incompletely separated crowns that have a **single root and root canal**.
- Cause unknown (trauma)/ hereditary tendency
- Gemination presents with **one tooth more** on counting.



GEMINATION



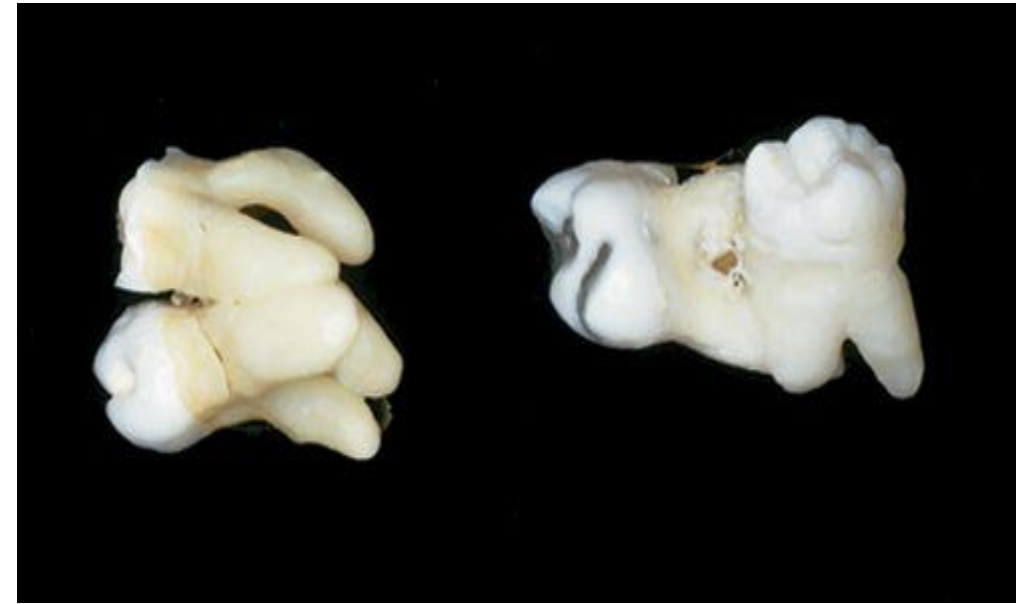
Twinning

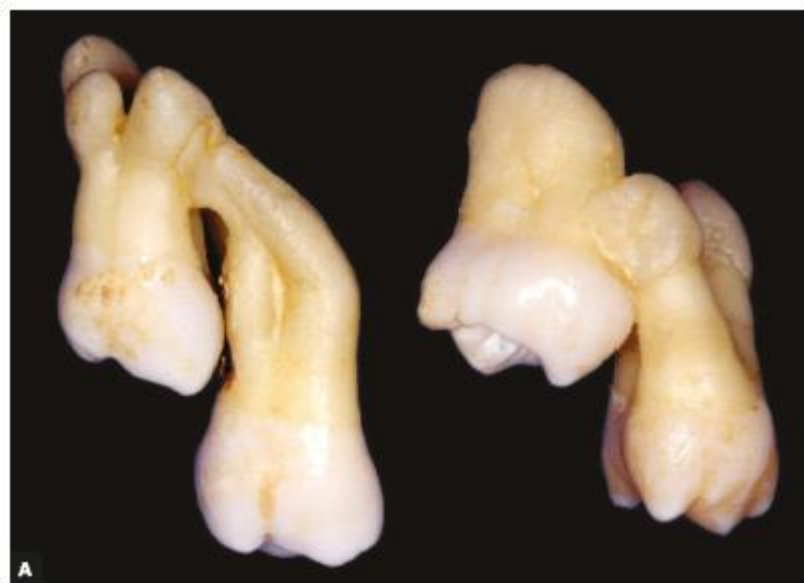


The term **'twinning'** has sometimes been used to designate the production of equivalent structures by division resulting in one normal and one supernumerary tooth.

CONCRESCENCE

- Concrecence is the joining of **tooth roots by cementum**.
- It occurs after root formation is complete.
- Cause: related to trauma or overcrowding
- **Clinical significance:** Extraction of one may result in extraction of the other.





Dilaceration

- **Dilaceration** is a severe bend in the long axis of the tooth.
- The bend is located at the junction between the crown and the root
- The bend can be as much as 90 degrees.
- Dilaceration usually results from trauma during development of root.
- **Clinical significance:** Difficulty at the time of extraction and difficulty in RCT.



TAURODONTISM

- **Taurodontism** literally means '**bull-like teeth**'.
Taurodont teeth are usually **molars**.
- They have elongated crowns and apically displaced furcation resulting in an abnormally **long pulpal chamber** and **shortened roots**.
- Clinically these teeth appear normal.
- Can be seen in association with **Down syndrome** and **Klinefelter's syndrome**.
- No specific treatment



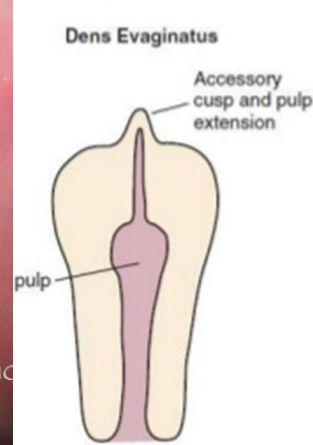
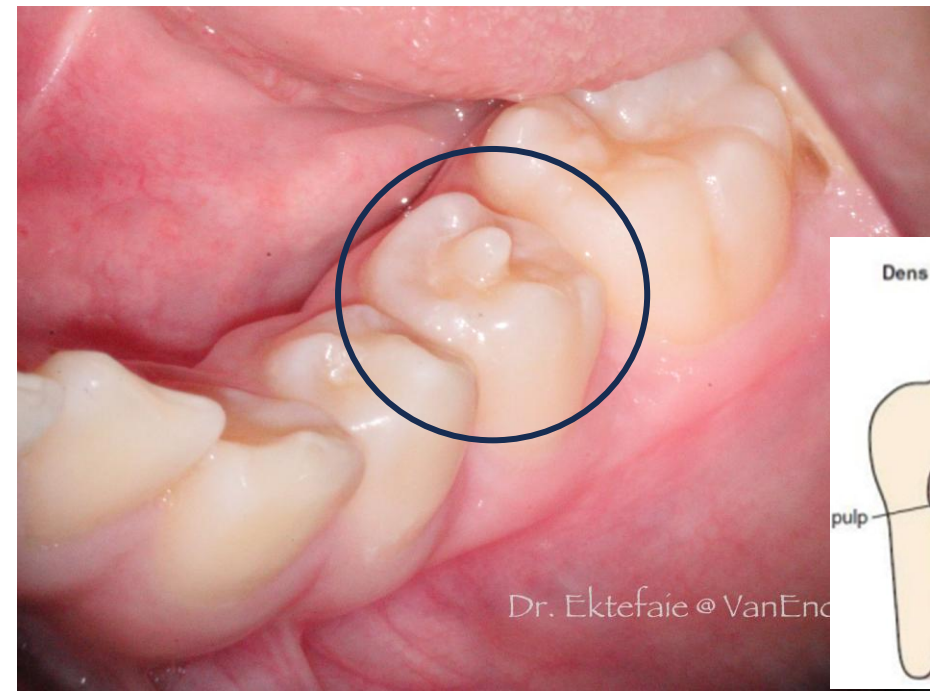
TALON CUSP

- A **talon cusp** is an extra cusp that resembles an eagle's talon
- A talon cusp appears as a projection from the cingulum of incisor teeth.
- These can interfere with occlusion and caries control.
- Prevalent in person with Rubinstein-Taybi syndrome.



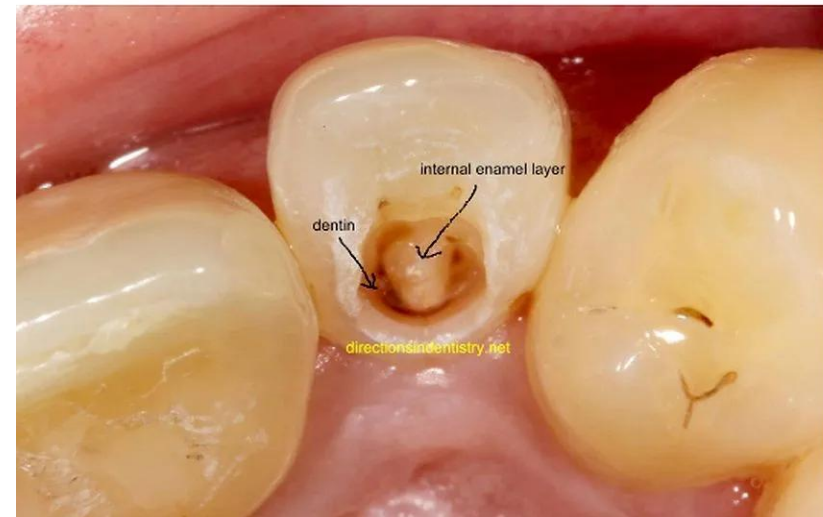
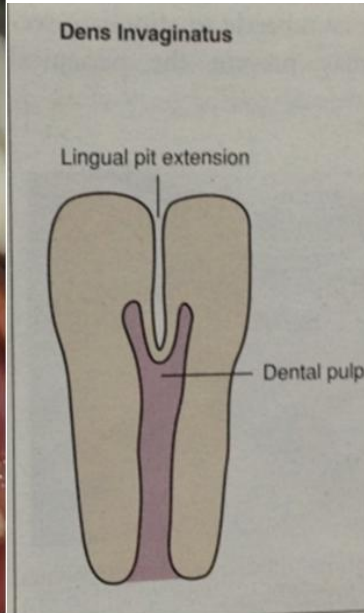
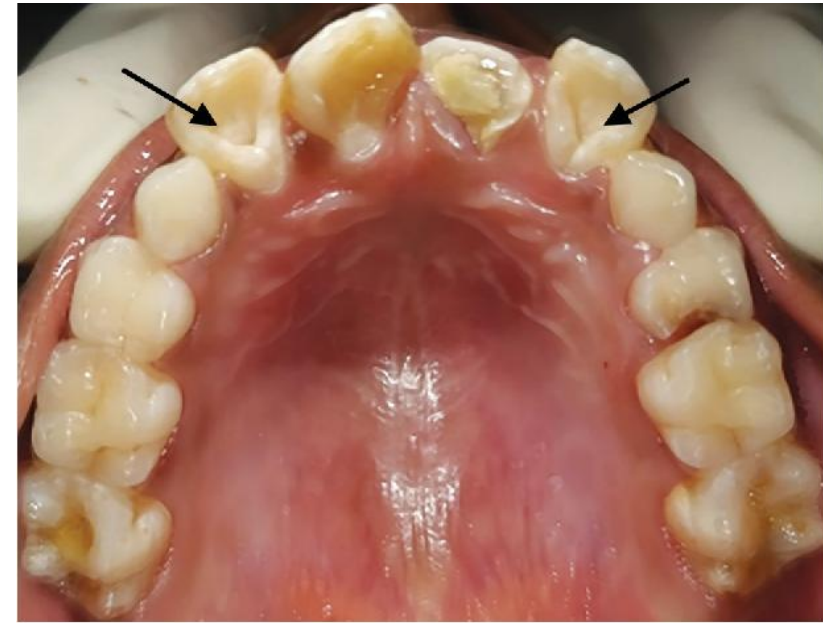
DENS EVAGINATUS

- It is projection of enamel can occur on the occlusal surface of the premolar teeth.
- This condition is also called as **Leong's premolar**/Occusal tuberculated odontome.
- Dens evaginatus can also **contain a pulp horn**.
- **Clinical significance:** incomplete eruption, displacement of teeth and/or pulp exposure with subsequent infection following occlusal wear or fracture.



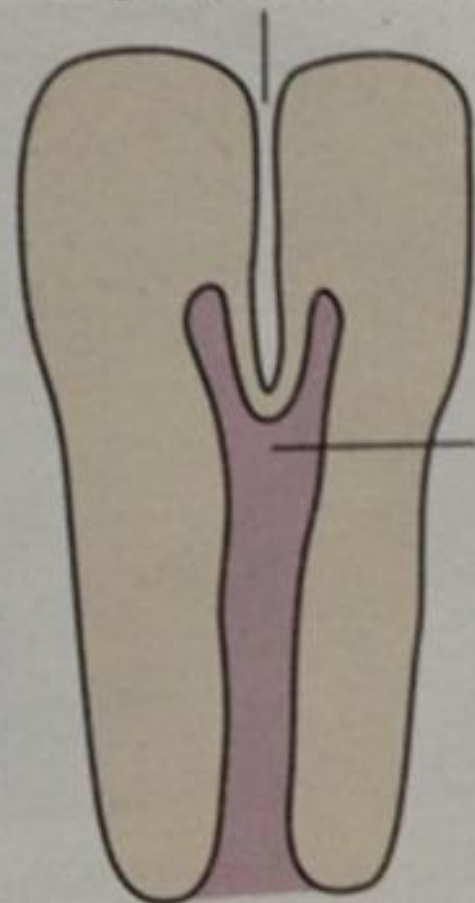
DENS IN DENTE

- **Dens in dente** means '**tooth within a tooth**' also known as **Dens invaginatus**.
- **Represents an exaggeration of lingual pit** that is lined by enamel.
- As a result of an invagination in the surface of a tooth crown before calcification has occurred.
- It is an uncommon: usually noted in **maxillary lateral incisors**.
- **Clinical importance:** To prevent caries, pulp infection, and premature loss of the tooth, the condition must be recognized early and the tooth prophylactically restored.



Dens Invaginatus

Lingual pit extension



Dens Evaginatus

Accessory cusp
and pulp
extension



Dental pulp

ENAMEL PEARLS

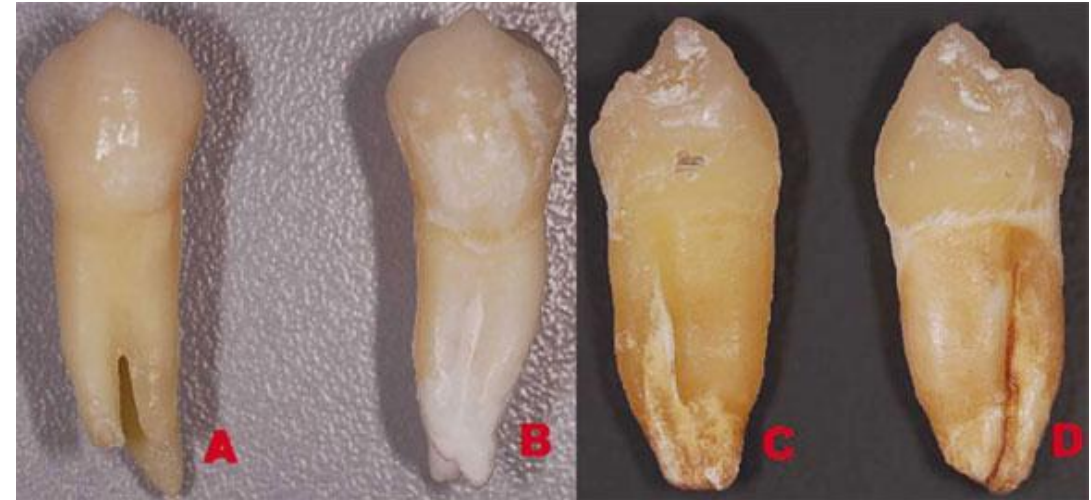
- **Enamel Pearls (enamelomas)** are small nodules of enamel found on the root of the tooth close to or at the cemento-enamel junction.



SUPERNUMERARY ROOT

Common in mandibular canines, premolars, and molars.

R/Fs are important when extractions and RCTs are necessary.



Formative assessment



THANK YOU

CASES OF DEVELOPMENTAL DISEASES AFFECTING NUMBER AND STRUCTURE OF TEETH

CASE DISCUSSION

SLO: Discuss the cases of the developmental diseases affecting structure and number of the Teeth.

CLO	LLO	By the end of this lecture, students should be able to:	
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Essential reading:

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Chapter 16 – Abnormalities of teeth

D/Disturbances affecting number of
teeth
(defect in initiation stage of odontogenesis)

ANODONTIA

- Absence of teeth.
- **Complete anodontia:** rare but associated with hereditary ectodermal dysplasia
- **Partial anodontia**(Hypodontia).
- **False anodontia** (result of extraction) &
- **Pseudo Anodontia** (unerupted teeth).
- **Pathogenesis:** failure of odontogenesis (initiation)
- T/t: Dental Prosthesis.



Missing lateral incisor





SUPERNUMERARY TEETH

- Extra teeth or Polydontia or hyperdontia.
- The most common of all supernumerary teeth is the **mesiodens** followed by **Fourth molars**, called '**paramolars**' or '**distomolars**'.
- Most commonly they are **impacted** hence radiographs help in diagnosis.

Etiology: Most accepted etiology: **Dental lamina hyperactivity**

Clinical Features

- Delayed or failure of eruption of permanent teeth
- Crowding and malalignment
- Diastema (especially midline)
- Rotation or displacement of adjacent teeth

Syndromes Associated with Supernumerary Teeth

- **Cleidocranial Dysplasia**
 - Multiple supernumerary teeth
 - Delayed eruption of permanent teeth
 - Hypoplastic or absent clavicles
- **Gardner's Syndrome**
 - Multiple impacted supernumerary teeth
 - Odontomas
 - Multiple osteomas of jaw
 - Multiple polyposis of the large intestine

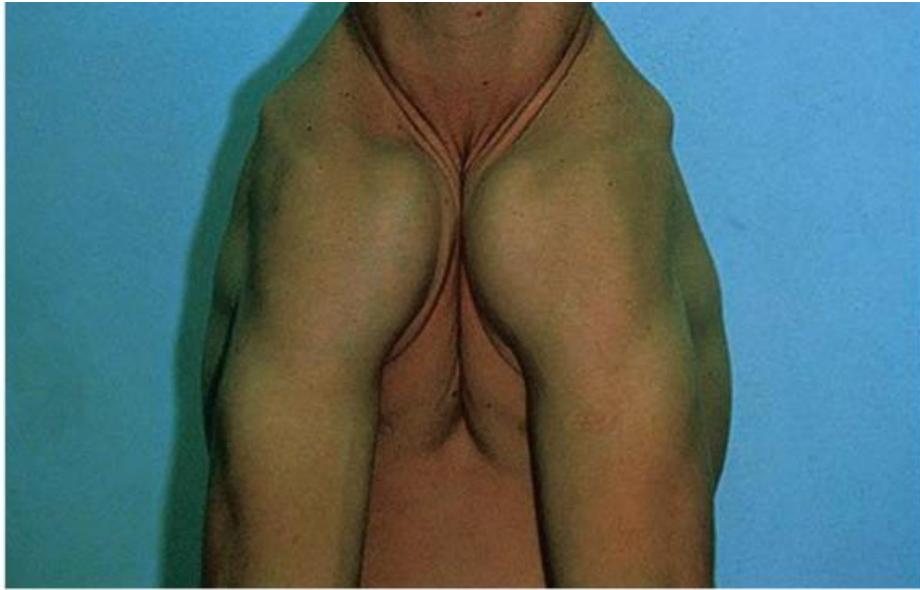


Fig. 20.5. **Cleidocranial dysplasia:** absent clavicles.

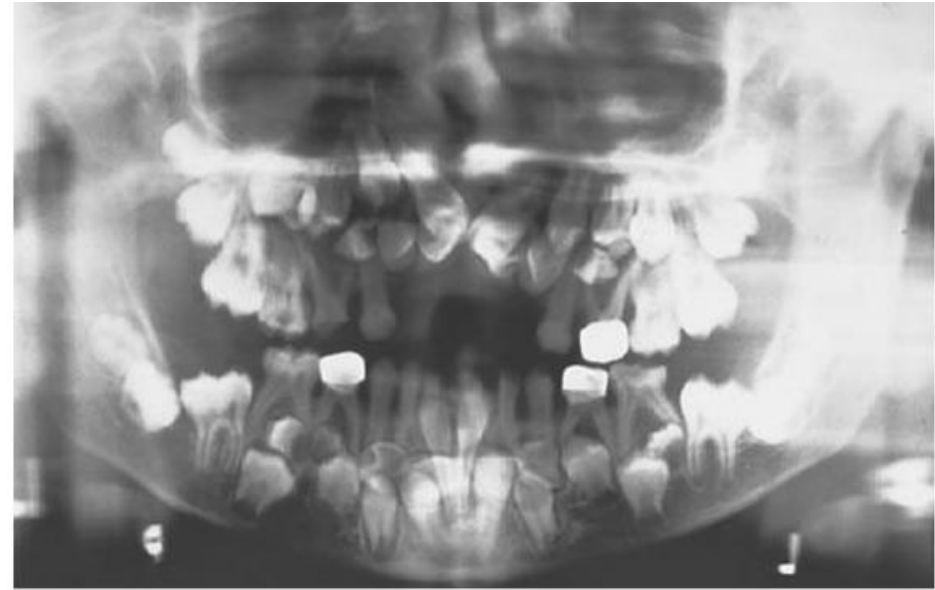


Fig. 20.6. **Cleidocranial dysplasia:** retained primary teeth.



Fig. 20.7. **Gardner syndrome:** mandibular osteomas.

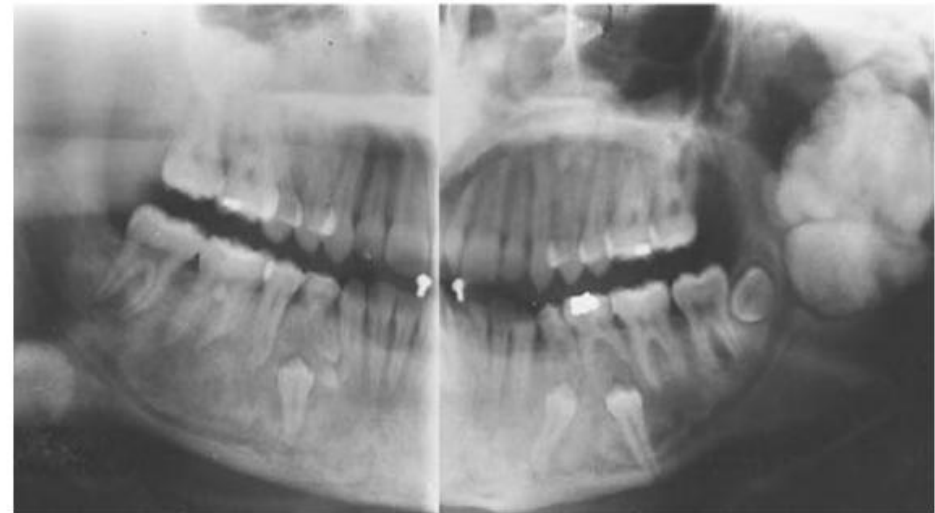


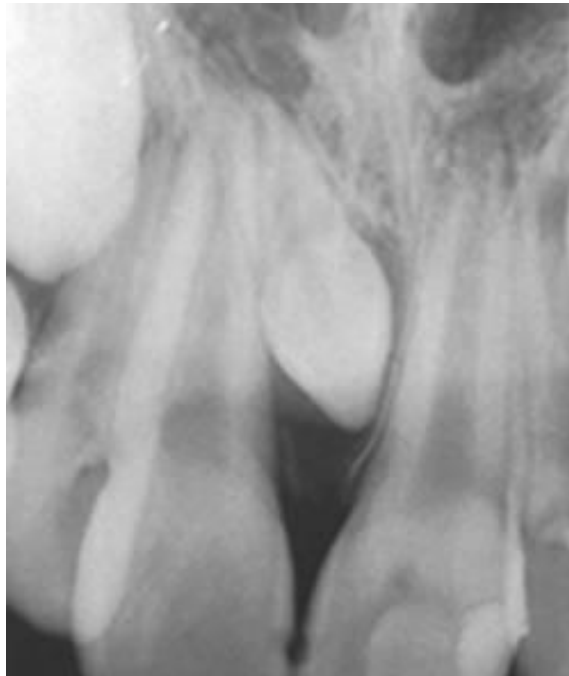
Fig. 20.8. **Gardner syndrome:** osteomas, supernumerary teeth.

Hyperdontia: Extra maxillary lateral incisor



Classification – Based on Location

- **Mesiodens** – between maxillary central incisors (*most common*)
- **Paramolar** – buccal or lingual to molars
- **Distomolar** – distal to third molar



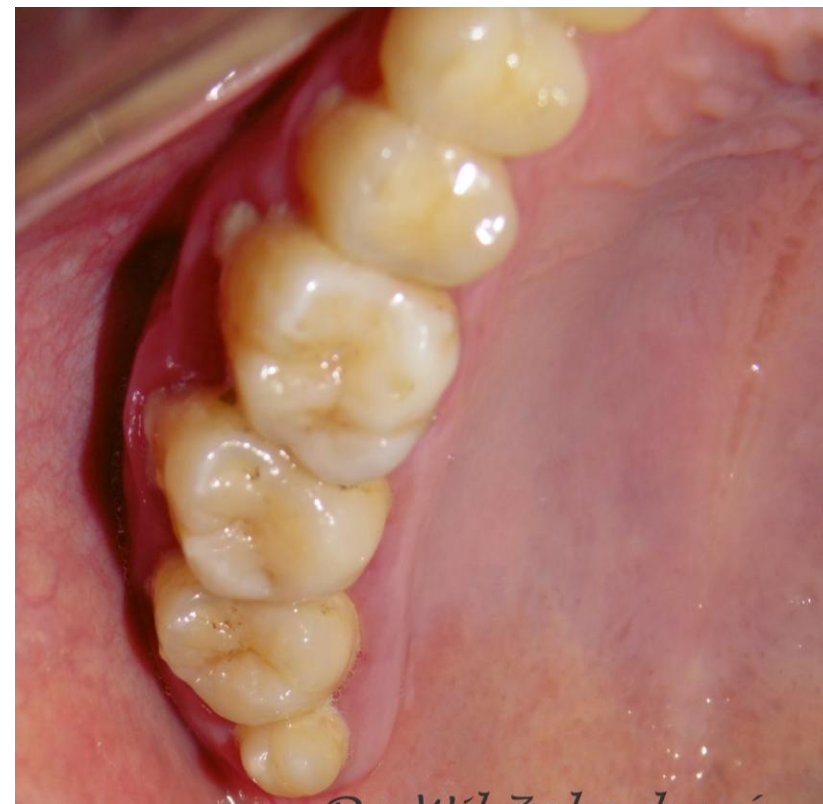
MESIODENS



PARAMOLAR



DISTOMOLAR



- Developmental diseases affecting structure of teeth

1. Amelogenesis Imperfecta
2. Dentinogenesis Imperfecta
3. Dentin Dysplasia
4. Regional Odontodysplasia

A healthy 14 years old female patient reported with a chief complaint of insufficient chewing functions due to hypersensitivity of the teeth and esthetic problems because of the colour of the teeth.



Diagnosis ?

AMELOGENESIS IMPERFECTA

(Enamel formation imperfect)

AMELOGENESIS IMPERFECTA

- A structural defect of the tooth enamel affecting both deciduous and permanent teeth.
- Amelogenins and non amelogenins are abnormal.
- **Genetics:**
 1. Autosomal dominant pattern
 2. Autosomal recessive pattern.
 3. X-linked form
 4. Other cases of this condition result from new gene mutations

Hypoplastic type

- Inadequate formation of enamel matrix.
- Pitted, grooved or translucent
- Radiographic features:
 - ✓ Reduced bulk of enamel
 - ✓ Dentin and pulp chamber normal



Figure 2-89 • Hypoplastic amelogenesis imperfecta, generalized pitted pattern. Note the numerous pinpoint pits scattered across the surface of the teeth. The enamel between the pits is of normal thickness, hardness, and coloration. (From Stewart RE, Prescott GH: *Oral facial genetics*. St Louis, 1976. Mosby.)

Hypo mineralized type

- Enamel matrix is normal in quantity but poorly calcified.



Figure 2-97 • Hypocalcified amelogenesis imperfecta. Dentition exhibiting diffuse yellow-brown discoloration. Note numerous teeth with loss of coronal enamel except for the cervical portion.



Hypomaturative type

- Enamel is normal in form on eruption but

- ✓ Opaque
- ✓ White to brownish-yellow
- ✓ Softer than normal
- ✓ Tends to chip from underlying dentin



Figure 2-95 • Hypomaturational amelogenesis imperfecta. Dentition exhibiting mottled, opaque white enamel with scattered brownish-yellow spots.



Figure 2-96 • Hypomaturational amelogenesis imperfecta, snowcapped pattern. Dentition exhibiting zone of white opaque enamel in the incisal and occlusal one fourth of the enamel surface. (Courtesy of Dr. Heddie O. Sedano.)

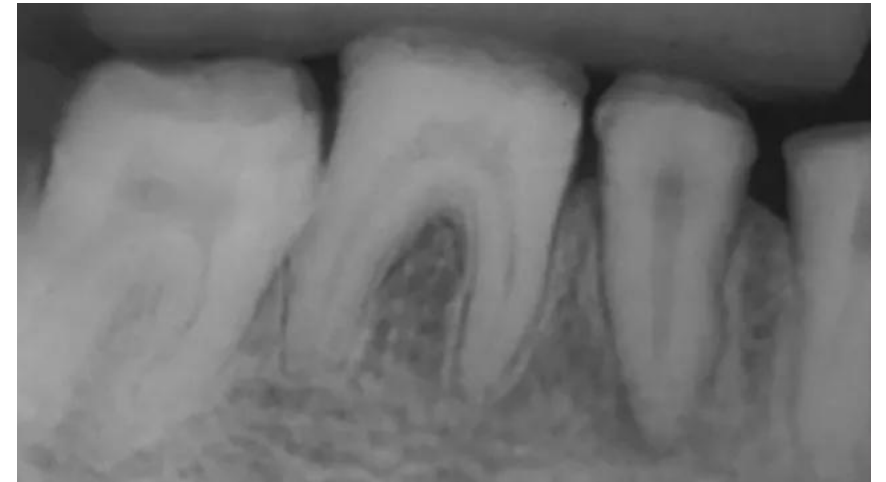
Clinical features:

- Abnormal color: yellow, brown or grey.
- Hypersensitive to temperature changes.
- Higher risk for dental cavities.

Radiographic features:

Treatment:

- Crowns -stainless steel crowns in children & porcelain-adults
- In the worst case scenario, the teeth may have to be extracted.





A 20 years old female, student by profession, reported with the chief complaint of rough and discoloured teeth.

Her secondary complaint was difficulty in chewing and sensitivity in teeth. She wanted to have white natural looking teeth with better chewing efficiency.

INTRAORAL EXAMINATION

- Chipped off incisal edges
- Generalized brown discoloration
- Plaque and calculus deposits
- Gingivitis



DIAGNOSIS AND TREATMENT

Amelogenesis Imperfecta (Hypoplastic type)

TREATMENT OPTIONS

- Veneers
- Full coverage crowns and FDP's



DENTINOGENESIS IMPERFECTA

Dentinogenesis imperfecta is an autosomal dominant condition affecting both deciduous and permanent teeth.

Hereditary Opalescent Dentin.

Sheilds classification

Type I: Dentinogenesis imperfecta seen **along with osteogenesis imperfecta.**

Type II: Occurs in people **without** Osteogenesis imperfecta.

Type III: known as **shell teeth or brandy wine type.**

Revised Sheilds classification:

Type I: Dentinogenesis imperfecta

Type II: known as **shell teeth or brandy wine type.**





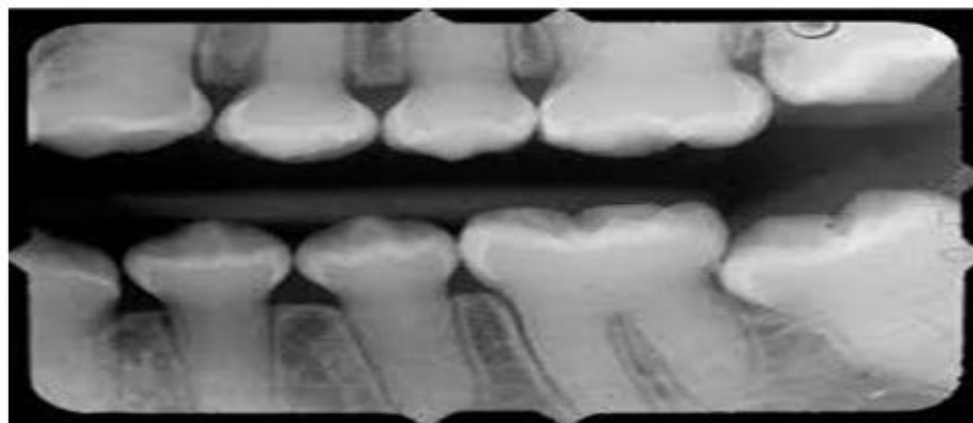
Figure 1. Pretreatment anterior view of 19 year-old service member with yellowish gray crowns, defective enamel and dentin, midline discrepancy and aesthetic disharmonies.

Dentinogenesis Imperfecta

BLUE
DISCOLORATION
OF
ENAMEL



- ☐ Bulbous crown
- ☐ Absent pulp chambers



Clinical features

- Discolored Teeth (most often a blue-gray or yellow-brown color) and translucent or opalescent hue.
- Teeth are also **weaker than normal**.

Radiographic features:

- Type I and II show **total obliteration of the pulp chamber**.
- Type III shows thin dentin and extremely enormous pulp chamber. Hence these teeth are known as **Shell Teeth**.

Histology:

- Dentinal tubules are **irregular and are bigger** in diameter. Areas of uncalcified matrix are seen. DEJ is flat.

Treatment:

- Similar to amelogenesis imperfecta mainly to provide **cosmetics to the patients**.
- Due to **abnormal pulpal anatomy the RCT is difficult** to perform.

A 10-year-old girl reported with malocclusion. Severe tooth wear had caused a loss of vertical dimension and insufficient occlusion.



DIAGNOSIS AND TREATMENT

- The primary disease was diagnosed as Dentinogenesis Imperfecta type II.



Dentin Dysplasia



TYPE- 1 ROOT LESS TEETH

DD Type I

Radicular dentin dysplasia

Characterized by:-

- 1.Both dentitions are affected.
- 2.Normal appearing crowns
- 3.No or only rudimentary root development
(rootless teeth)
- 4.Incomplete or total obliteration of the pulp chamber.
- 5.Teeth may exhibit extreme mobility and exfoliate prematurely.





DD type II

coronal dentin dysplasia

Characterized by:-

1. partial pulpal obliteration.
2. **Thistle-tube**-or **flame-shaped** coronal pulp chambers
3. Thread-like root canals
4. Usually the **absence** of periapical radiolucencies.
5. In this type of anomaly, **teeth roots are of normal shape and contour.**



TYPE – 2 (THISTLE TUBE PULP)



Histopathologically

The enamel and the immediately subjacent dentin appear normal.

Deeper layers of dentin show an **atypical tubular pattern with an amorphous, atubular area, and irregular organization.**

Normal dentinal tubule formation appears to have been blocked so that new dentine forms around obstacles and takes on the characteristic appearances described as “**lava flowing around boulders**”



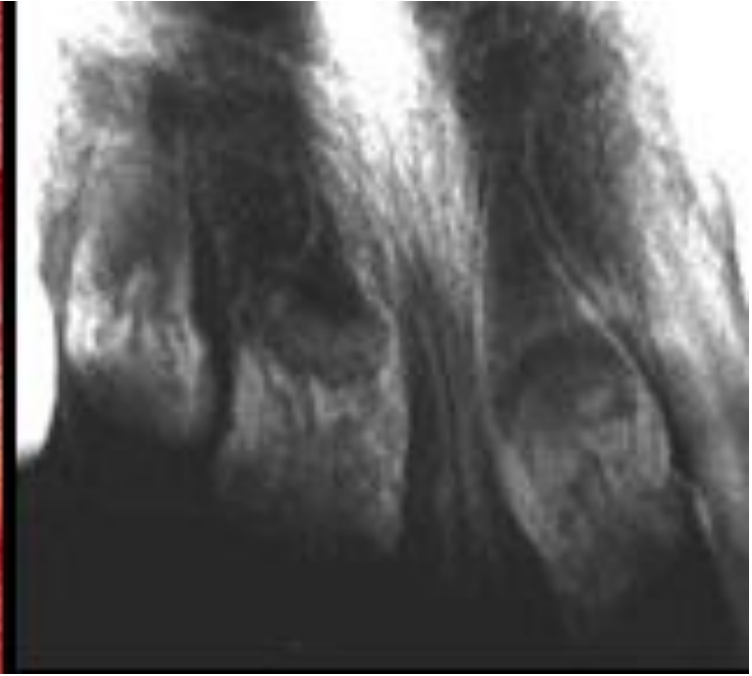
A 12-year-old boy was referred to Dental Clinics due to excessive painful swelling of his mandibular left cheek region. The crowns of his teeth had normal morphologic characteristics, but the color of his teeth was slightly more yellow than expected. The maxillary and mandibular central and lateral incisors were mobile.



DD type I, confirming the diagnosis based on the clinical and radiographic features.

REGIONAL ODONTODYSPLASIA

- It usually affects **localized area of teeth & it is nonhereditary**.
- The enamel, dentin, and pulp of teeth are affected, and on radiographs the teeth are described as "**ghost teeth**".
- **Mostly idopathic** but some are associated with syndromes, growth abnormalities, neural disorders, and vascular malformations.



Case



- Panoramic and periapical radiographs showed longitudinal shortened teeth with lack of contrast between enamel and dentine. The affected teeth had wide pulp chambers with open apices and were smaller than normal in overall size. The main radiographic feature was a 'ghost' appearance.



THANK YOU!!!!

KINDLY REFER

**Regezi and Sciubba
(Oral Pathology Clinico-pathologic Correlation)**